Extension: From Incompressible to Compressible Navier--Stokes Equations

1. Motivation

The incompressible Navier--Stokes equations assume that the fluid density is constant:

$$\rho = \text{constant}.$$

Under this assumption, mass conservation reduces to the divergence-free condition

$$\nabla \cdot \mathbf{u} = 0.$$

This is appropriate for many low-speed liquid flows and low-Mach-number gas flows.

However, in compressible flow, density may vary in space and time:

$$\rho = \rho(x, y, t).$$

This occurs in flows involving:

- high-speed gases,
- shock waves,
- acoustic waves,
- strong pressure variations,
- significant temperature variations.

Therefore, the incompressibility constraint must be replaced by a full conservation law for mass.

2. From Incompressibility to Compressibility

For incompressible flow, the continuity equation is

$$\nabla \cdot \mathbf{u} = 0.$$

For compressible flow, the continuity equation becomes

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

In two dimensions, this is

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0.$$

This equation states that changes in density are caused by the net flux of mass into or out of a control volume.

The velocity field is no longer required to satisfy

$$\nabla \cdot \mathbf{u} = 0.$$

Instead, compression and expansion are allowed.

3. Compressible Momentum Equations

For compressible flow, the momentum equations are most naturally written in conservation form.

In two dimensions, the conservation of x-momentum is

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} = -\frac{\partial p}{\partial x} + \text{viscous terms.}$$

The conservation of y-momentum is

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} = -\frac{\partial p}{\partial y} + \text{viscous terms.}$$

The key difference from the incompressible case is that momentum is now expressed as

$$\rho u, \qquad \rho v,$$

rather than simply u and v.

Thus, density is dynamically coupled to the velocity field.

4. Need for an Energy Equation

In incompressible flow, pressure is often obtained from a Poisson equation that enforces

$$\nabla \cdot \mathbf{u} = 0.$$

In compressible flow, pressure is not usually obtained from a pressure Poisson equation.

Instead, pressure is coupled to density and temperature through thermodynamics.

For this reason, compressible flow requires an additional conservation law: the energy equation.

A common conservative form is

$$\frac{\partial E}{\partial t} + \nabla \cdot ((E + p)\mathbf{u}) = \text{viscous and heat-conduction terms,}$$

where E is the total energy per unit volume.

For an inviscid compressible flow, the total energy is

$$E = \rho e + \frac{1}{2}\rho(u^2 + v^2),$$

where:

- ρe is the internal energy per unit volume,
- $\frac{1}{2}\rho(u^2 + v^2)$ is the kinetic energy per unit volume.

5. Equation of State

The compressible system requires a relationship between pressure, density, and temperature.

For an ideal gas, the equation of state is

$$p = \rho R T,$$

where:

- p is pressure,
- ρ is density,
- R is the gas constant,
- T is temperature.

For a calorically perfect gas, one may also write

$$p = (\gamma - 1)\rho e,$$

where:

- e is the internal energy per unit mass,
- γ is the ratio of specific heats.

This equation closes the system by linking the thermodynamic variables.

6. Compressible Euler Equations

Before adding viscosity, it is useful to study the inviscid compressible equations, called the Euler equations.

In two dimensions, the conservative variables are

$$\mathbf{q} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}.$$

The compressible Euler equations can be written as

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{q})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{q})}{\partial y} = 0,$$

where $\mathbf{F}$ and $\mathbf{G}$ are flux vectors in the x- and y-directions.

The flux in the x-direction is

$$\mathbf{F}(\mathbf{q}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E+p) \end{bmatrix}.$$

The flux in the y-direction is

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E+p) \end{bmatrix}.$$

This conservative form is especially important because compressible flows may contain discontinuities such as shocks.

7. Compressible Navier--Stokes Equations

The compressible Navier--Stokes equations extend the Euler equations by including viscous stresses and heat conduction.

They can be written schematically as

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \cdot \mathbf{F}_{\text{inviscid}}(\mathbf{q}) = \nabla \cdot \mathbf{F}_{\text{viscous}}(\mathbf{q}, \nabla \mathbf{q}).$$

The conserved variables remain

$$\mathbf{q} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}.$$

Compared with incompressible Navier--Stokes, the major changes are:

- density is unknown and evolves in time,
- pressure is obtained from an equation of state,
- an energy equation is required,
- shock waves and expansion waves may appear,
- conservative numerical methods become essential.

8. Mach Number

Compressibility effects are often measured using the Mach number:

$$M = \frac{|\mathbf{u}|}{a},$$

where:

- $|\mathbf{u}|$ is the flow speed,

- a is the speed of sound.

For an ideal gas,

$$a = \sqrt{\gamma R T}.$$

Interpretation:

- $M \ll 1$: nearly incompressible flow,
- $M < 1$: subsonic compressible flow,
- $M = 1$: sonic flow,
- $M > 1$: supersonic flow.

Compressibility effects become increasingly important as the Mach number increases.

9. Main Conceptual Differences

The move from incompressible to compressible flow changes the mathematical structure of the problem.

Feature	Incompressible Flow	Compressible Flow
Density	Constant	Variable
Continuity equation	$\nabla \cdot \mathbf{u} = 0$	$\rho_t + \nabla \cdot (\rho \mathbf{u}) = 0$
Pressure	Constraint variable	Thermodynamic variable
Pressure equation	Often Poisson equation	Equation of state
Energy equation	Often not required	Required
Main unknowns	u, v, p	$\rho, \rho u, \rho v, E$
Shocks	Not usually present	Possible
Numerical form	Often primitive variables	Conservative variables

10. Implementation Roadmap for Compressible Flow

The implementation pathway differs from the incompressible solver.

Step 1: Define the conserved variables

Use

$$\mathbf{q} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}.$$

These are the variables advanced in time.

Step 2: Recover primitive variables

From the conserved variables, compute

$$u = \frac{\rho u}{\rho}, \qquad v = \frac{\rho v}{\rho}.$$

The kinetic energy per unit volume is

$$K = \frac{1}{2}\rho(u^2 + v^2).$$

The internal energy contribution is

$$E - K.$$

For an ideal gas,

$$p = (\gamma - 1)\left[E - \frac{1}{2}\rho(u^2 + v^2)\right].$$

Step 3: Construct fluxes

Compute the inviscid flux vectors

$$\mathbf{F}(\mathbf{q}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(E + p) \end{bmatrix},$$

and

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(E + p) \end{bmatrix}.$$

Step 4: Advance the solution in time

Use a conservative finite-volume or finite-difference update of the form

$$\mathbf{q}_{i,j}^{n+1} = \mathbf{q}_{i,j}^{n} - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1/2,j} - \mathbf{F}_{i-1/2,j}\right) - \frac{\Delta t}{\Delta y}\left(\mathbf{G}_{i,j+1/2} - \mathbf{G}_{i,j-1/2}\right).$$

The quantities $\mathbf{F}_{i+1/2,j}$ and $\mathbf{G}_{i,j+1/2}$ are numerical fluxes at cell interfaces.

Step 5: Apply boundary conditions

Boundary conditions must be imposed on the conserved or primitive variables.

Typical compressible-flow boundary conditions include:

- inflow,
- outflow,
- slip wall,
- no-slip wall,
- reflective wall,
- far-field boundary.

The choice depends on the physical problem.

Step 6: Choose a stable time step

For compressible flow, the time step is constrained by the fastest wave speed:

$$\Delta t \leq C_{\mathrm{CFL}} \min \left(\frac{\Delta x}{|u| + a}, \frac{\Delta y}{|v| + a} \right),$$

where a is the speed of sound and C_{CFL} is a chosen CFL number.

This differs from incompressible flow because acoustic waves must also be resolved.

Step 7: Monitor physical admissibility

The solution must remain physically meaningful:

$$\rho > 0, \qquad p > 0.$$

Negative density or pressure usually indicates numerical instability or an unsuitable discretisation.

11. Recommended Next Model Problem

Before attempting the full compressible Navier--Stokes equations, it is advisable to study the compressible Euler equations in one dimension.

A standard first model problem is the shock-tube problem.

The one-dimensional Euler equations are

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix} + \frac{\partial}{\partial x} \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix} = 0.$$

The shock-tube problem begins with a discontinuity between two constant states:

$$(\rho_L, u_L, p_L) \qquad \text{and} \qquad (\rho_R, u_R, p_R).$$

This problem produces:

- a shock wave,
- a contact discontinuity,
- a rarefaction wave.

It is a natural first test for compressible-flow solvers.

12. Summary

The move from incompressible to compressible flow introduces several major changes.

In incompressible flow:

$$\rho = \text{constant}, \qquad \nabla \cdot \mathbf{u} = 0.$$

Pressure acts mainly as a constraint variable enforcing incompressibility.

In compressible flow:

$$\rho = \rho(x, y, t),$$

and mass conservation becomes

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

The unknowns become

$$\rho, \qquad \rho u, \qquad \rho v, \qquad E.$$

Pressure is obtained from an equation of state rather than a pressure Poisson equation.